**Question 1: Calculating Bandwidth Utilization**
An auditor is assessing a web server's bandwidth utilization. The server has a 1 Gbps (gigabit per second) network interface. During peak hours, the server handles 450,000 HTTP requests per hour, with each request averaging 500 KB in response size. Calculate the percentage of bandwidth utilized during peak hours.
**Solution:**

1. **Calculate Total Data Transferred per Hour:**
   - Each request transfers 500 KB.
   - Total data per hour = 450,000 requests/hour × 500 KB/request = 225,000,000 KB/hour.
2. **Convert KB to Gigabits:**
   - 1 KB = 8 kilobits; therefore, 225,000,000 KB = 1,800,000,000 kilobits.
   - 1 Gbps = 1,000,000,000 kilobits per second.
   - Total data per hour in gigabits = 1,800,000,000 kilobits / 1,000,000 = 1,800 gigabits.
3. **Calculate Bandwidth Utilization:**
   - 1 hour = 3,600 seconds.
   - Bandwidth utilized per second = 1,800 gigabits / 3,600 seconds = 0.5 Gbps.
   - Percentage utilization = (0.5 Gbps / 1 Gbps) × 100% = 50%.

**Answer:** The web server utilizes 50% of its bandwidth during peak hours.

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**Question 2: Estimating Storage Requirements for Log Files**
A web application generates log entries of 1 KB for each user transaction. If the application processes an average of 2 million transactions daily, estimate the monthly storage requirement for the log files.
**Solution:**

1. **Daily Log Generation:**
   - 2 million transactions/day × 1 KB/transaction = 2,000,000 KB/day.
2. **Convert KB to GB:**
   - 1 GB = 1,000,000 KB.
   - Daily log size = 2,000,000 KB / 1,000,000 = 2 GB/day.
3. **Monthly Storage Requirement:**
   - Assuming a 30-day month:
   - 2 GB/day × 30 days = 60 GB.

**Answer:** The application requires approximately 60 GB of storage for log files each month.

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**Question 3: Calculating SSL/TLS Overhead**
An e-commerce site is considering implementing SSL/TLS to secure transactions. SSL/TLS encryption adds an overhead of approximately 5% to the data size. If the average page size is 800 KB, what will be the new average page size after implementing SSL/TLS?
**Solution:**

1. **Calculate Overhead:**
   - 5% of 800 KB = 0.05 × 800 KB = 40 KB.
2. **New Page Size:**
   - 800 KB + 40 KB = 840 KB.

**Answer:** The new average page size will be 840 KB after implementing SSL/TLS.

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**Question 4: Evaluating Server Load Capacity**
A web server has a maximum capacity of handling 10,000 simultaneous connections. During an audit, it's observed that the server experiences a 20% increase in connections every year. If the current peak usage is 6,000 simultaneous connections, in how many years will the server reach its maximum capacity?
**Solution:**

1. **Define Variables:**

- Current connections ($C_0$) = 6,000.
- Growth rate (r) = 20% per year = 0.20.
- Maximum capacity ($C_m$) = 10,000.

2. **Apply Growth Formula:**
   - $C_n = C_0 \times (1 + r)^n$, where $C_n$ is the number of connections after n years.
   - $10,000 = 6,000 \times (1.20)^n$.
3. **Solve for n:**
   - $(1.20)^n = 10,000 / 6,000 \approx 1.6667$.
   - Taking the logarithm:
     - $n \times \log(1.20) = \log(1.6667)$.
     - $n = \log(1.6667) / \log(1.20) \approx 2.74$.

**Answer:** The server will reach its maximum capacity in approximately 2.74 years, or roughly 2 years and 9 months.

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**Question 5: Assessing Firewall Throughput Requirements**

An organization plans to implement a firewall to protect its web servers. The combined inbound and outbound traffic is estimated at 3 terabytes (TB) per day. The firewall should handle at least 25% more than the average daily traffic to accommodate traffic spikes. What is the minimum throughput capacity the firewall should support in Gbps?

**Solution:**

1. **Calculate Total Daily Traffic with Overhead:**
   - 25% of 3 TB = 0.25 × 3 TB = 0.75 TB.
   - Total = 3 TB + 0.75 TB = 3.75 TB/day.
2. **Convert TB to Terabits:**
   - 1 TB = 8 Terabits
   - 3.75 TB = 3.75 × 8 = 30 Terabits per day
3. **Convert to Bits per Second:**
   - 1 day = 86,400 seconds
   - Required throughput = 30 Terabits / 86,400 seconds
   - $= 30 \times 10^{12}$ bits / 86,400 seconds
   - = 347,222,222 bits per second ≈ 347 Mbps
4. **Convert to Gbps:**
   - 347 Mbps = 0.347 Gbps

**Answer:**

The firewall should support a minimum throughput of **0.35 Gbps** to handle daily traffic, including a 25% overhead for traffic spikes.

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## Question 6: Web Server Uptime Calculation

A company has a web server with a guaranteed uptime of 99.9% as per the SLA (Service Level Agreement). Calculate the maximum downtime (in minutes) the server can have per month.

**Solution:**

1. **Total Minutes in a Month:**
   - Assuming 30 days in a month:
   - Total minutes per month = 30 × 24 × 60 = **43,200 minutes**
2. **Allowed Downtime:**
   - SLA guarantees 99.9% uptime → Downtime = (100% - 99.9%) = 0.1%
   - Allowed downtime = (0.1/100) × 43,200 = **43.2 minutes**

**Answer:**

The web server can have a maximum **downtime of 43.2 minutes per month** under a 99.9% uptime SLA.

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## Question 7: Web Server Request Handling Capacity

A web server processes requests at an average speed of **150 milliseconds per request**. If the server operates 24/7, how many requests can it handle in a **single day**?

**Solution:**

1. **Total Seconds in a Day:**
   - 24 hours × 60 minutes × 60 seconds = **86,400 seconds**
2. **Requests Per Second (RPS):**
   - 150 milliseconds = 0.15 seconds per request
   - Requests per second = 1 / 0.15 = **6.67 requests per second**
3. **Total Requests Per Day:**
   - 86,400 seconds × 6.67 requests/second ≈ **576,000 requests/day**

**Answer:**

The web server can handle approximately **576,000 requests per day**.

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## Question 8: Storage Requirement for Web Server Logs

A web server generates logs at a rate of **20 KB per request**. If the server processes **500,000 requests per day**, how much storage (in GB) is needed for one year of logs?

**Solution:**

1. **Daily Log Storage:**
   - 500,000 requests × 20 KB/request = **10,000,000 KB/day**
   - Convert to GB: **10,000,000 KB ÷ 1,000,000 = 10 GB/day**
2. **Annual Storage Requirement:**
   - 10 GB/day × 365 days = **3,650 GB**
   - Convert to TB: **3,650 GB ÷ 1,000 = 3.65 TB**

**Answer:**

The web server requires **3.65 TB of storage per year** for logs.

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## Question 9: Load Balancing Distribution Across Web Servers

A company uses **5 load-balanced web servers** to handle an average of **2 million requests per hour**. If one server fails, how many additional requests per second must each remaining server handle?

**Solution:**

1. **Total Requests Per Hour:**
   - 2,000,000 requests/hour
   - Convert to requests per second: **2,000,000 ÷ 3,600 = 555.56 RPS**
2. **Requests Per Server Before Failure:**
   - 555.56 RPS ÷ 5 servers = **111.11 RPS per server**
3. **Requests Per Server After One Fails (Now 4 Servers):**
   - 555.56 RPS ÷ 4 servers = **138.89 RPS per server**
4. **Additional Load Per Server:**
   - 138.89 RPS - 111.11 RPS = **27.78 RPS per server**

**Answer:**

Each remaining server must handle an additional **27.78 requests per second** if one server fails.

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## Question 10: Power Consumption of Web Servers

A data center operates **20 web servers**, each consuming **450 watts per hour**. If electricity costs **$0.12 per kWh**, calculate the total electricity cost for running these servers for **one month (30 days)**.

**Solution:**

1. **Power Consumption Per Server Per Day:**
   - 450 watts × 24 hours = **10,800 watt-hours = 10.8 kWh**

2. **Total Power Consumption for 20 Servers Per Day:**
    - 10.8 kWh × 20 servers = **216 kWh/day**
3. **Monthly Power Consumption:**
    - 216 kWh/day × 30 days = **6,480 kWh**
4. **Total Electricity Cost:**
    - 6,480 kWh × $0.12/kWh = **$777.60**

**Answer:**
The total electricity cost for running 20 web servers for a month is **$777.60**.

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